

Abstract Algebra II - Homework 3

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(a)

Let $f(x) = c_mx^m + c_{m-1}x^{m-1} + \cdots + c_1x + c_0$, $c_i \in K \forall i \in \{0, 1, \dots, m\}$. Let $\sigma \in G$ and $a_i \in X$, we have: (Clearly all elements in G fix K and $f(a_i) = 0$ and $\sigma(0) = 0$)

$$\begin{aligned}\sigma(f(a_i)) &= \sigma(c_ma_i^m) + \sigma(c_{m-1}a_i^{m-1}) + \cdots + \sigma(c_1a_i) + \sigma(c_0) \\ &= c_m(\sigma(a_i))^m + c_{m-1}(\sigma(a_i))^{m-1} + \cdots + c_0 \\ &= f(\sigma(a_i)) = 0\end{aligned}$$

Thus, $\sigma(a_i)$ is also a root of f , so $\sigma(a_i) \in X$, since σ is a bijection on the field L , its restriction to the finite set X is a bijection from X to itself. Hence, every $\sigma \in G$ defines a permutation of X .

Define $\phi : G \rightarrow \text{Perm}(X)$, $\sigma \in G$ by $\phi(\sigma) = \sigma|_X$ (the restriction of σ to the set X). For any $\sigma, \tau \in G$ and any $a \in X$, we have:

$$\phi(\sigma \circ \tau)(a) = (\sigma \circ \tau)(a) = \sigma(\tau(a)) = \phi(\sigma)(\phi(\tau)(a))$$

It's obviously a homomorphism. We then try to prove the map is injective, i.e., the kernel of ϕ is the identity automorphism. By definition, the splitting field L is the smallest field containing K and all the roots of f , therefore, L is generated by the roots over K , in other words, $L = K(a_1, a_2, \dots, a_n)$.

Suppose $g \in \ker(\phi)$, any field automorphism in G is completely determined by its action on the generators of the field extension. Since g fixes K and fixes every generator a_i , it must fix every element of L , thus, g is the identity map. Because $\ker \phi = \text{id}_L$, the map ϕ is injective.

(b)

Theorem 1.1 (Isomorphism extension theorem). *Given any field F , an algebraic extension field E of F and an isomorphism φ mapping F onto a field F' then φ can be extended to an isomorphism τ mapping E onto an algebraic extension E' of F' (a subfield of the algebraic closure of F').*

Theorem 1.2. *Let F be a field of characteristic 0 and let $f(x) \in F[x]$, and let G be the Galois group of $f(x)$ over F . If $f(x)$ is irreducible, G acts transitively on the roots of $f(x)$.*

Proof. Because f is an irreducible polynomial over K , for any two roots $a_i, a_j \in X$, there exists a field isomorphism between the simple extensions:

$$\theta : K(a_i) \rightarrow K(a_j)$$

that fixes K and maps a_i to a_j . Because L is the splitting field of f over both $K(a_i)$ and $K(a_j)$, this isomorphism θ can be extended to an automorphism σ of the entire splitting field L (by the Isomorphism Extension Theorem).

Therefore, there exists some $\sigma \in G$ such that $\sigma(a_i) = a_j$. This means the action of G on the set of roots X is transitive (there is only one orbit). $\square$

By the orbit-stabilizer theorem, G acts on set X , the size of the orbit of an element multiplied by the size of its stabilizer subgroup equals the size of the group:

$$|G| = |\text{Orbit}(a_i)| |\text{Stab}_G(a_i)|$$

Since the action is transitive, $|\text{Orbit}(a_i)| = |X| = n$, hence n divides $|G|$.

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Theorem 2.3. *Symmetric group S_p (where p is a prime) can be generated by a transposition and a p -cycle.*

Proof. Consider any transposition $\tau = (ab)$, and the p -cycle $\sigma = (12 \dots p)$. Because $\langle \sigma \rangle$ is the subgroup of S_p consists of p -cycles, and every element has order p (p is prime) and can be the generator. Hence, there exists integer k such that σ^k sends a to b and is a p -cycle. $\square$

To makes the notation easier, we can relabel the elements such that the position of a is marked as 1, and the position of $\sigma^k(a) = b$ is marked as 2 and so on. So the problem becomes finding the group generated by $(12, d \dots p)$ and (12) . Since they can generate all kinds of adjacent transpositions, all transposition can be made by using adjacent transposition, all permutation can be made by muliple transpositions. Thus, (12) and $(12 \dots p)$ generate S_p .

Let the Galois group of f be G .

By taking the derivative, $f'(x) = 5x^4 - 5 = 5(x-1)(x+1)(x^2+1)$, the local minimum of f is $f(1) = -5$ and the local maximum is $f(-1) = 3$. By some calculation we found: $f(0) = -1$, $\lim_{x \rightarrow \infty} f(x) = \infty$, $\lim_{x \rightarrow -\infty} f(x) = -\infty$. So $f(\infty)f(1) < 0$, $f(0)f(-1) < 0$, $f(-1)f(-\infty) < 0$, by intermediate value theorem, there exists at least one real roots on these intervals respectively: $(-\infty, -1)$, $(-1, 0)$, $(1, \infty)$, because the problem states that the real roots are less than 5, and

the complex roots must be in conjugation forms (i.e. in pairs), so there are 3 real roots and 2 complex roots.

Let the complex roots be z and $\bar{z}$. The automorphism corresponding to complex conjugation fixes the 3 real roots and swaps z and $\bar{z}$. This is a transposition in G .

Since the G acts transitively on the roots of f , and since G is Galois, G is isomorphic to a subgroup of S_5 . To maintain the group structure, we need to find transitive subgroup of S_5 .

Consider a more general case, the galois group permutes p roots with p being prime. By orbit-stabilizer theorem, $|G|$ is divided by p . By cauchy's theorem, it contains a element of order p , which corresponds to the p -cycle in S_p . Here, we found G contains a 5-cycle.

By Theorem 2.3, since G contains both 5-cycle and a transposition, the only corresponding subgroup of S_5 is itself. As a result, the Galois group of f is isomorphic to S_5 .